



## Li-Fi: Tomorrow's Connectivity With The Power Of Light



Philips Lighting's Li-Fi device

► **The visible light spectrum remains an untapped resource with a large bandwidth, making Li-Fi the connectivity technology of tomorrow**

**A**s more and more devices get connected to the Internet, soon there will be a need for bigger bandwidth to accommodate the massive volume of data transactions. Light

through the Ethernet cable to be coded into light packets. Emitted light is modulated at a high frequency (not detectable by human eye) to carry data. A small device containing sen-

sors and microprocessors is required, which reads and decodes the light packet and displays the data on the screen.

Fidelity (Li-Fi) can be a stepping stone towards a new era of communication that utilises visible light. Although in developmental stages, Li-Fi offers a great scope of applications in big business areas like banks, large enterprises, aviation and even defence.

### How Li-Fi works

The underlying technology of Li-Fi involves a sophisticated technical upgrade of luminaires that enable seamless transfer of data through the visible light spectrum. The luminaire consists of a normal RJ-45 Ethernet port (also present in Wi-Fi routers), through which the Ethernet cable is connected.

Inside the luminaire there is a small modem and some specific components that allow the data being brought

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### Scopes and benefits of Li-Fi

Currently, various pilots and proofs-of-concept of Li-Fi projects have been deployed, mainly in large enterprises and certain government sectors.

Data travelling through light cannot be hacked. It can be accessed only by those who are available within the coverage range of the luminaire. Usually, a luminaire has three-metre coverage radius, ensuring only a handful of personnel can access the data contained in the light. This improves the control on access to the network. Since Wi-Fi usually works in the radio frequency (RF) bandwidth (2.4GHz and 5.0GHz), the risk of signal interference becomes nil using Li-Fi (430THz-770THz).

Sumit Joshi, vice chairman and managing director, Philips Lighting India, says, "Li-Fi is significantly better than Wi-Fi in applications where RFs may interfere with the equipment, such as in hospitals, or places where Wi-Fi signals cannot reach or are weak, such as underground locations, or secure bank rooms handling sensitive data. Additionally, it can provide secure, personal connectivity for senior management or anyone else concerned about data privacy." Defence and aviation industries are also leveraging Li-Fi.

There are more benefits. Deepak Solanki, founder and chief executive